



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF PHYSICAL SCIENCES

PHYSICAL SCIENCES

GRADE 12

— CHEMISTRY—

18 March 2024

Examiner: I Pelser

Moderator: Dr L Whiting

TIME: 1 hour

MARKS: 50

INSTRUCTIONS AND INFORMATION

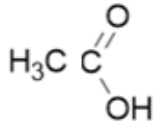
1. This question paper consists of 4 questions.
2. Answer ALL the questions in the ANSWER BOOK.
3. Start EACH question on a NEW page in the ANSWER BOOK.
4. Number the answers correctly according to the numbering system used in this question paper.
5. You may use a non-programmable calculator. You may use appropriate mathematical instruments.
6. NO data sheets are needed.
7. Give brief motivations, discussions et cetera where required.
8. Write neatly and legibly.

Question 1**[10]**

1.1 The EMPIRICAL formula of hexanoic acid is:

- A $\text{C}_3\text{H}_6\text{O}$
- B $\text{C}_3\text{H}_6\text{O}_2$
- D $\text{C}_6\text{H}_6\text{O}_2$
- D $\text{C}_6\text{H}_{12}\text{O}_2$

1.2 The structures of four organic compounds are shown below.

COMPOUND 1	COMPOUND 2
	$\text{H}_2\text{C CH}_2$
COMPOUND 3	COMPOUND 4
$\text{H}_3\text{C CHCH}_2$	$\text{H}_3\text{C CH}_2\text{ CH}_3$

Which of these compounds will decolourise bromine water FAST?

- A 1 only
- B 1, 2 and 3
- C 2 and 3
- D 1 and 4

1.3 Which of the following compounds has the highest vapour pressure?

- A CH_3CH_3
- B CH_3OCH_3
- C $\text{CH}_3\text{CH}_2\text{OH}$
- D CH_3COOH

- 1.4 Esters are formed by the reaction between two organic compounds, X and Y, each with a different functional group. The functional groups of these compounds are:

	Compound X	Compound Y
A	Hydroxyl group	Carboxyl group
B	Hydroxyl group	Carbonyl group
C	Hydroxide ion	Carboxyl group
D	Hydroxide ion	Carbonyl group

- 1.5 When butane is subjected to high temperatures and pressures, the following reaction takes place:



Which ONE of the following represents Y?

- A CHCCH_3
- B CH_2CHCH_3
- C $\text{CH}_3\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CHCHCH}_3$

Question 2

[19]

A to F in the table below represent six organic compounds

A	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$	B	Pentanal
C	Methyl butanoate	D	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{CH}-\text{CH}_2-\text{C} \\ \quad \quad \quad \quad \\ \quad \quad \quad \quad \text{O} \\ \quad \quad \quad \quad \\ \quad \quad \quad \quad \text{OH} \end{array}$
E	$\begin{array}{c} \text{O} \\ // \\ \text{H}_3\text{C}-\text{C} \\ \\ \text{CH}_2-\text{CH}_2-\text{CH}_3 \end{array}$	F	$\text{CH}_3(\text{CH}_2)_3\text{CH}_3$

2.1 Consider the organic compound A.

2.1.1 Is it a PRIMARY, SECONDARY or TERTIARY alcohol? Give a reason for the answer. (2)

2.1.2 Write down the IUPAC name for the above-mentioned compound. (2)

2.1.3 The above-mentioned compound undergoes an elimination reaction. Write down the STRUCTURAL FORMULA of the major product that is formed. (2)

2.2 Consider the organic compound B.

2.2.1 Define the term functional isomer. (2)

Write down the:

2.2.2 Homologous series to which compound B belongs (1)

2.2.3 IUPAC name of its functional isomer (2)

2.2.4 Letter that represents an isomer of Compound B (1)

2.3 Write down the balanced equation using MOLECULAR FORMULAE for the complete combustion of compound F. (3)

2.4 Name compound D (2)

2.5 Draw the structural formula of C (2)

Question 3**[10]**

- 3.1 The vapour pressures of two organic compounds at a specific temperature are compared.

	COMPOUND	MOLAR MASS (g·mol ⁻¹)
A	Butane	58
B	Methylpropane	58

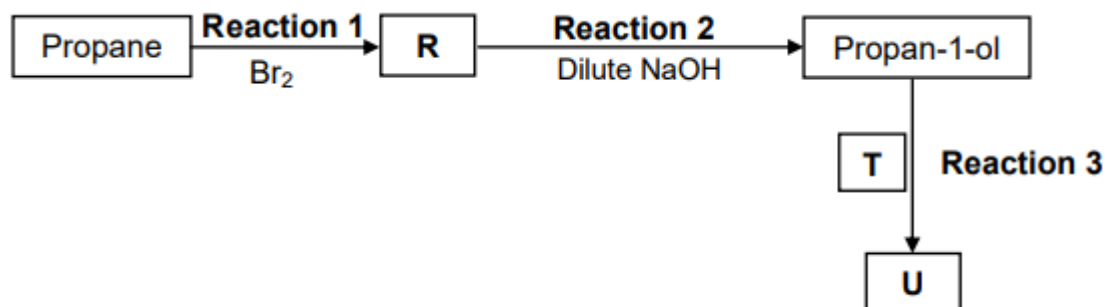
- 3.1.1 Define the term vapour pressure. (2)
- 3.1.2 Which ONE of the two compounds has the highest vapour pressure?
Write down only A or B. (1)
- 3.1.3 Fully explain the answer to QUESTION 3.1.2 (3)
- 3.1.4 Give a reason why the above comparison can be considered fair. (1)
- 3.2 The table below shows the results of an investigation regarding the melting points of organic molecules

	Organic compound	Boiling point (°C)
A	2,2-dimethylbutane	50
B	2,3-dimethylbutane	58
C	3-methylpentane	63

- 3.2.1 Write down the independent variable for this investigation. (1)
- 3.2.2 Explain the trend in boiling points of the above investigation. (2)

Question 4**[11]**

The flow diagram below shows various organic reactions using propane as starting reactant.



R, T and U represent different organic compounds.

Compound T is a CARBOXYLIC ACID and compound U is a FUNCTIONAL ISOMER of pentanoic acid.

Write down the NAME of the type of reaction represented by:

4.1 Reaction 1 (1)

4.2 Reaction 2 (1)

Consider reaction 1 and reaction 2.

4.3 Write down the IUPAC name of compound R. (2)

Reaction 3 takes place in the presence of a catalyst and heat.

Write down the:

4.4 IUPAC name of compound T (2)

4.5 NAME of compound U (2)

4.6 Write down a balanced equation in molecular formula to represent reaction 2 (3)

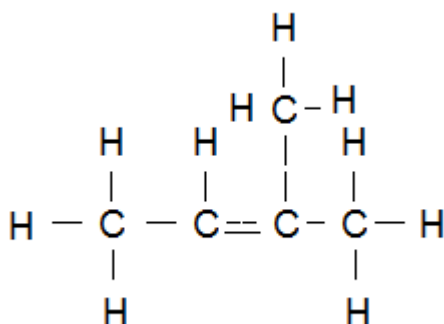
Memo

- 1.1 D
- 1.2 C
- 1.3 A
- 1.4 A
- 1.5 B

Question 2

2.1.1 Tertiary The functional group is bonded unto a carbon that is bonded unto 3 other carbons

2.1.2 2 - methylbutan – 2 – ol



2.1.3

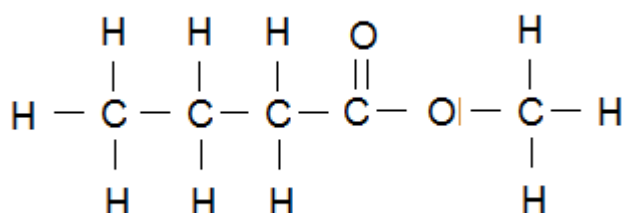
2.2.1 Compounds with the same molecular formula but belong to different homologous series

2.2.2 aldehyde

2.3 $\text{C}_5\text{H}_{12} + 8\text{O}_2 \rightarrow 5\text{CO}_2 + 6\text{H}_2\text{O}$

2.4 E

2.5



Question 3

- 3.1.1 The pressure a vapour exerts when the vapour and liquid are in equilibrium in a closed system
- 3.1.2 B
- 3.1.3 Methylpropane has a smaller surface area than butane, less London forces between the molecules, less energy needed to break the bonds, higher vapour pressure
- 3.1.4 Only TWO variables, surface area and vapour pressure, mass is controlled and the same for both

- 3.2.1 position of the side chains and chain length
- 3.2.2 A is an asymmetrical, compound B has more symmetry and C is a symmetrical compound with side chain in the middle. More asymmetrical, less bonds form, less energy needed to break the bonds, lower BP

Question 4

- 4.1 substitution/halogenation/bromination
- 4.2 substitution/hydrolysis
- 4.3 2 – bromopropane
- 4.4 ethanoic acid
- 4.5 propyl ethanoate
- 4.6 $\text{C}_3\text{H}_7\text{Br} + \text{NaOH} \rightarrow \text{C}_3\text{H}_7\text{OH} + \text{NaBr}$